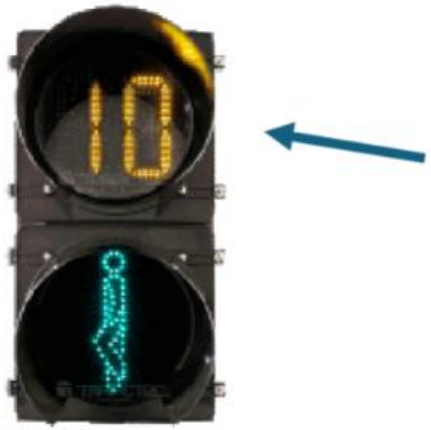
Lab 6

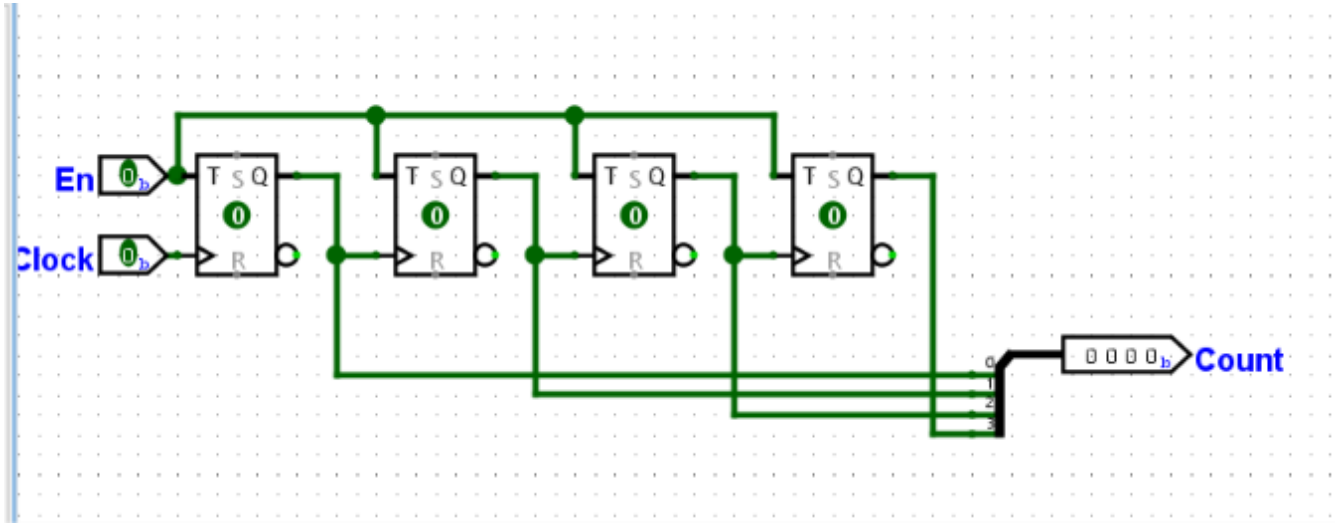


Goal is to build a register that down counts from 10 to 0
It should have a reset button to restart the countdown from 10

It will be incorporated with the bin2Hex converter and a 7 segment display

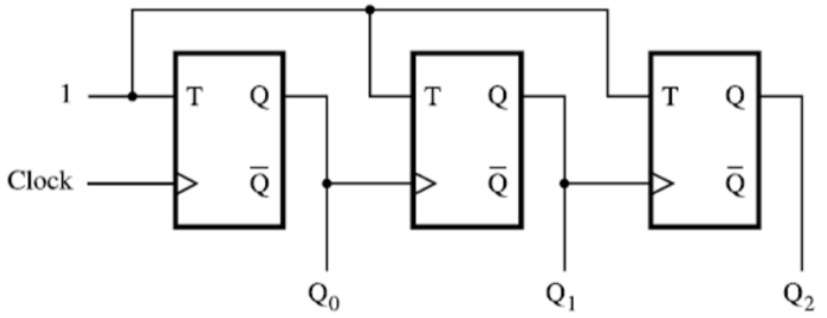
Helpful Registers

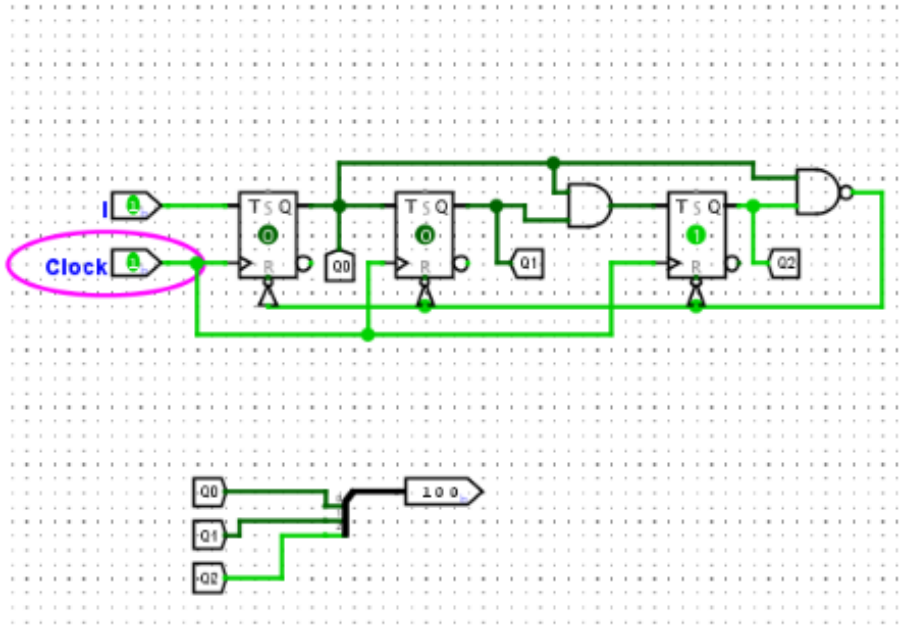
Modulus & Down Counters



Down counters, count down from its maximum value 2^n to 0. and automatically resets and repeats the sequence

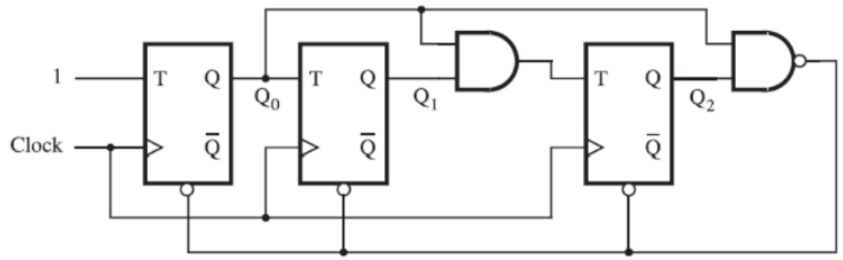
Down Counter Template





MOD Counters use sequential logic to count a certain number of states before resetting to its initial state

Modulo-5 Counter Template



(a) Circuit

MOD 5 will count, 0,1,2,3,4,5,0...

This is the range of number we want (10-0) in binary

Q3	Q2	Q1	Q0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

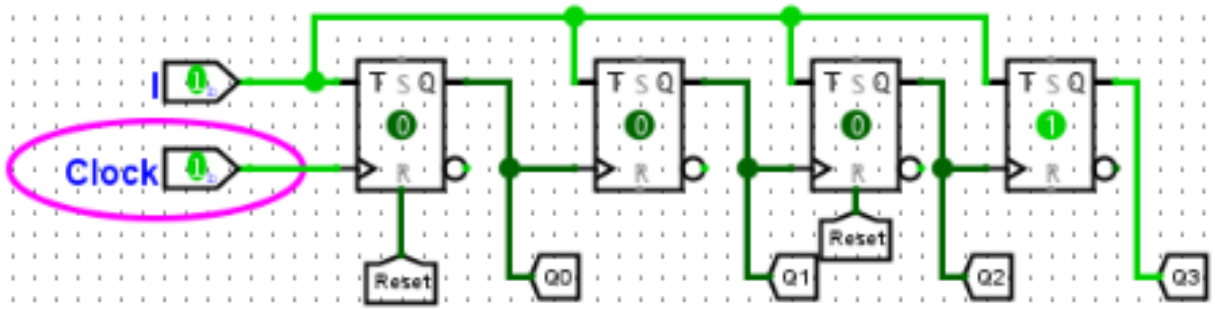


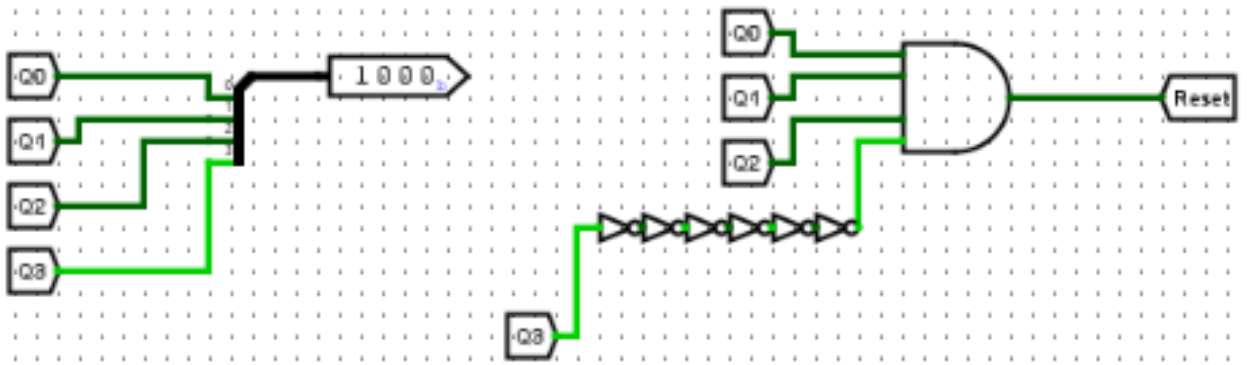
A 4 bit down counter will count from 15-0 and then rest to 15. but the numbers 15-11 are not desired in this situation. So, when the counter resets to 15. It makes sense to skip to 10 and continue down counting from there.

15 (D) = 1111 (B)

So when Q0&Q1&Q2&Q3==0

I want to reset pins Q0 and Q2 yielding 1010 (B) = 10 (D)
And continue down counting from there.





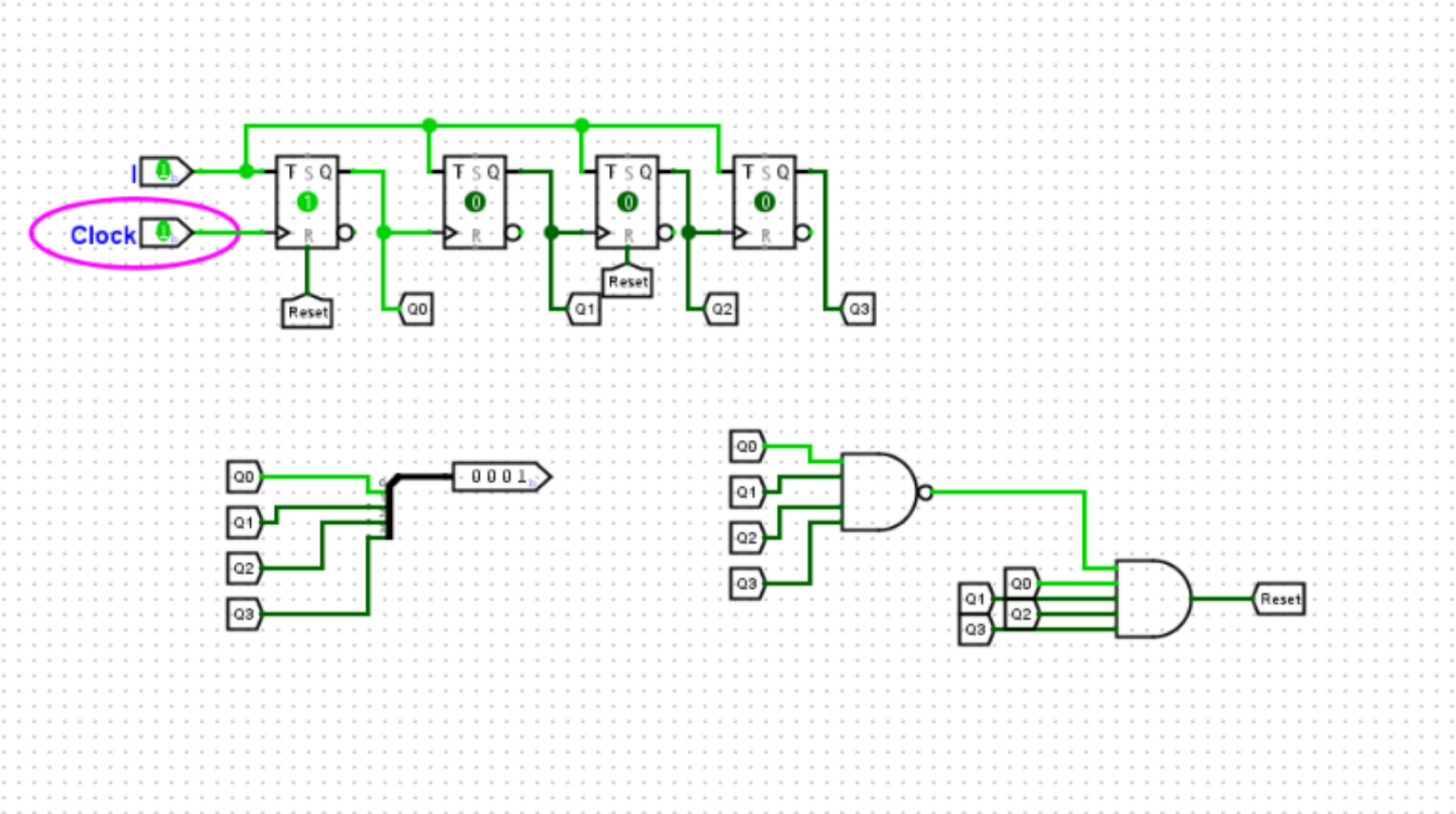
So I built this circuit

But it had an issue,

When switching from 8 to 7. Momentarlity all the pins would be high, triggering a reset and messing up the counter.

It would go 10,9,8,2,1,0,10…

I tried adding inverters as a buffer, but it didn’t work.



Then I built this, which also didn't work. Maybe I am looking at something wrong, but it almost seems like Q3 was existing in 2 states at the same time